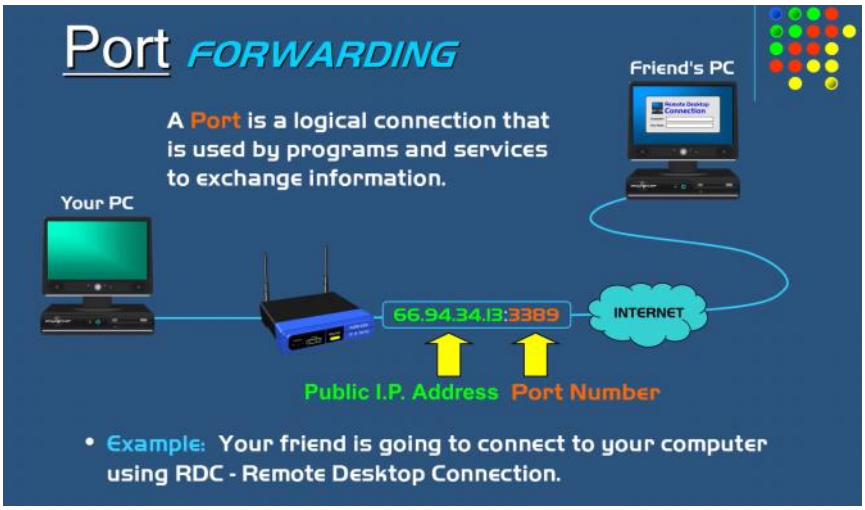
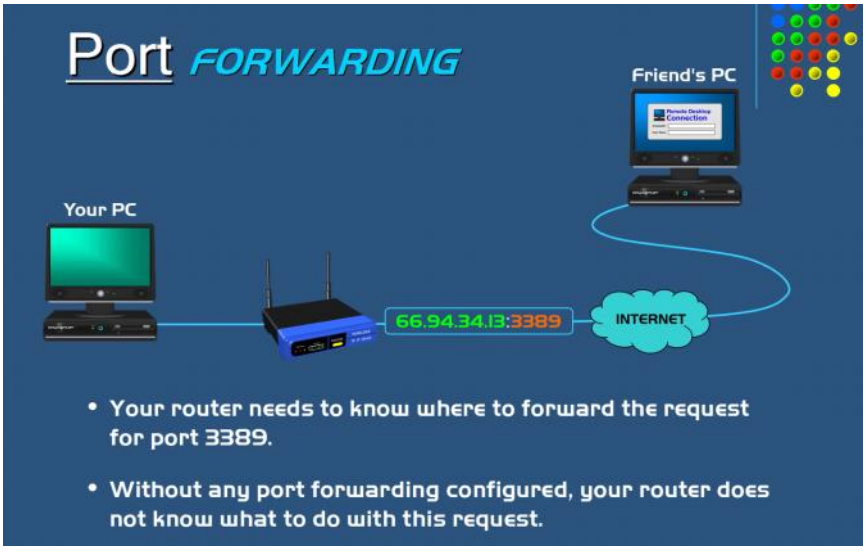
# Port Forwarding

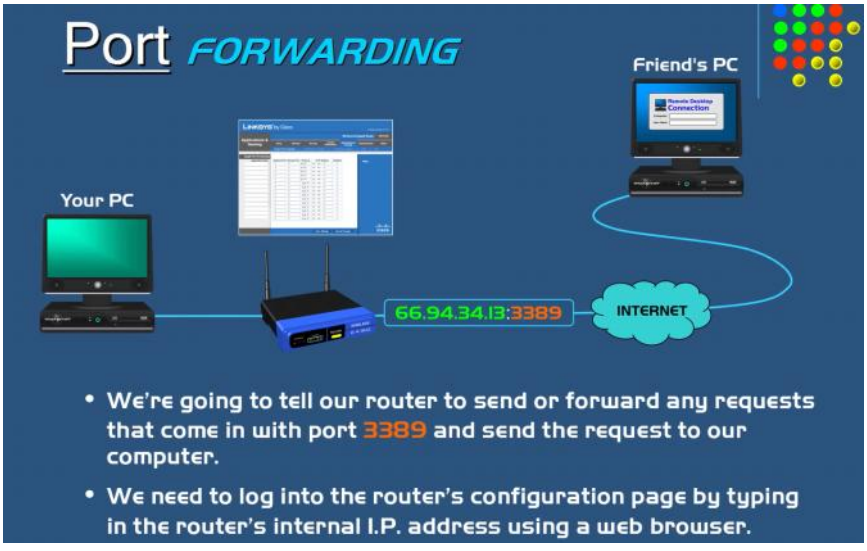
Thursday, March 5, 2026 3:09 PM

Port forwarding allows computers on the internet to connect to a specific computer or service within a private network, essentially making a computer accessible over the internet even when behind a router. This is achieved by configuring the router to forward requests coming in on a specific port number to a designated internal IP address.

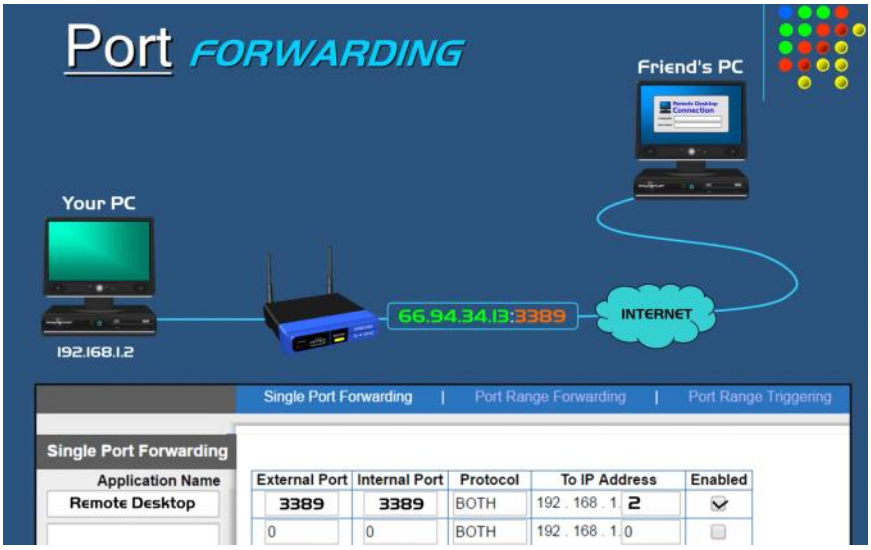


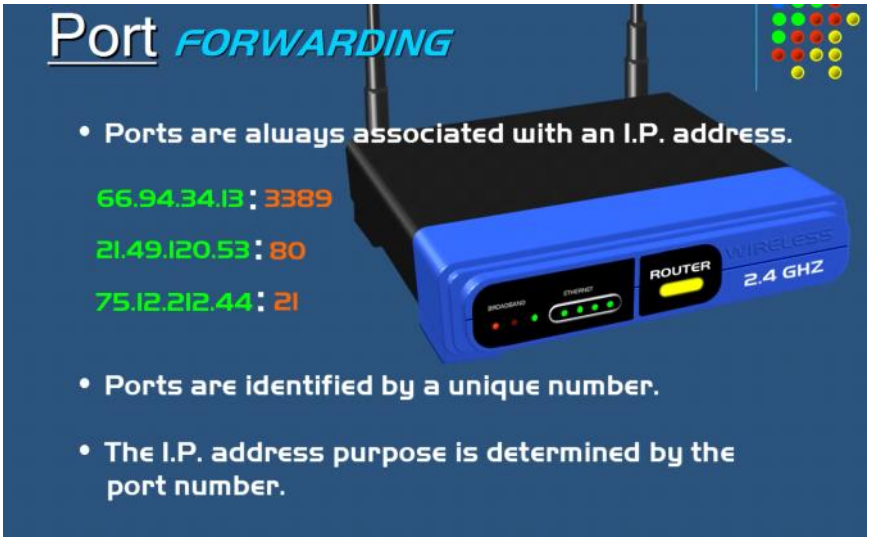

Ports are logical connections used by programs and services to exchange information, uniquely identifying these services. For example, Remote Desktop Connection uses port 3389. The process involves logging into the router's configuration page and setting up a rule that directs incoming requests for a particular port to the internal IP address of the target computer.

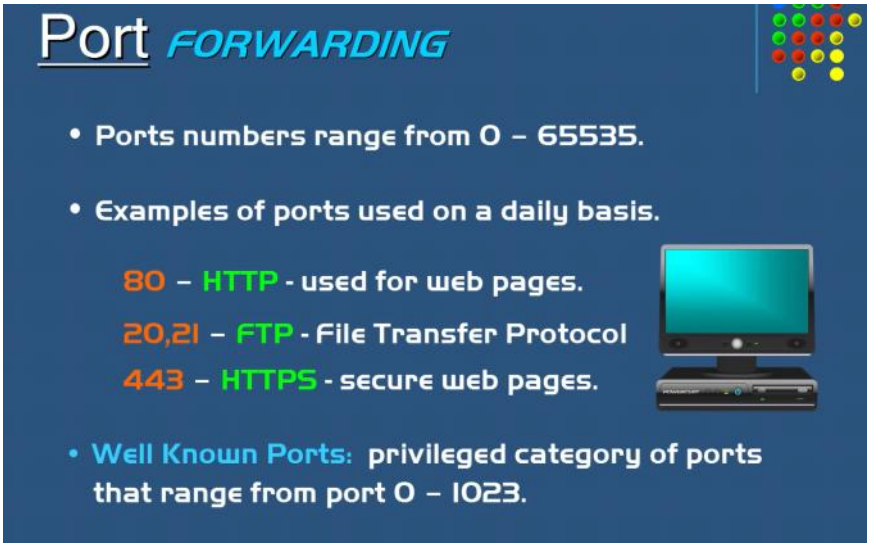


Port FORWARDING
LINKSYS by Cisco
Setup
Wireless-N Gigabit Router
ROUTER
2.4 GHZ


Port FORWARDING
Friend's PC
Your PC
192.168.1.2
66.94.34.13:3389
INTERNET
Single Port Forwarding | Port Range Forwarding | Port Range Triggering
Single Port Forwarding
Application Name
Remote Desktop
External Port
Internal Port
Protocol
To IP Address
Enabled
3389
3389
BOTH
192 . 168 . 1 . 2
0
0
BOTH
192 . 168 . 1 . 0


Port FORWARDING
• Ports are always associated with an I.P. address.
66.94.34.13 : 3389
21.49.120.53 : 80
75.12.212.44 : 21
ROUTER
2.4 GHZ
• Ports are identified by a unique number.
• The I.P. address purpose is determined by the port number.




The concept is similar to an old-fashioned phone switchboard: the router acts like the phone, the IP address is like the main phone number bringing the call to the company, and the port number is like an extension number that directs the call to a specific department (or computer/service). Ports are always associated with an IP address and an application or process, determining the purpose of the connection (e.g., web pages, FTP). Port numbers range from 0 to 65535, with some common ones like Port 80 for web pages and Port 443 for secure web pages. The well-known ports range from 0 to 1023.